

Results

Candidate Outcome

Candidate Outcome I

The generated loop selected exp-mlp-tanh-64 (One-hidden-layer tanh MLP, MLP) after evaluating 4 candidate(s) from a proposed set of 5. The nearest_centroid_baseline (nearest-centroid) baseline reached 82.6% test_accuracy, while the selected candidate reached 89.4%, an absolute change of 6.8%. The selected model has 50890 parameters. The transformer-candidate evaluated flag is true, and the candidate budget exhausted flag is true, which means the ledger records 1 deferred proposal(s) rather than expanding the run automatically.

Candidate Outcome I

The benchmark score is 1. That score is not a model-quality claim by itself; it is a compact grading artifact for the methods contract: metric improvement, budget compliance, offline execution, and selected-candidate recording. Rank-stability diagnostics report that the selected candidate is top ranked in 72.5% of deterministic bootstrap resamples, with runner-up `exp-mlp-relu-32`. The candidate score figure is registered as fig. 1, the confusion matrix as fig. 3, the per-class diagnostic as fig. 4, the learning curves as fig. 5, the complexity diagnostic as fig. 7, the selected-candidate error examples as fig. 8, the candidate lifecycle diagnostic as fig. 9, rank

Candidate Outcome I

stability as fig. 2, the training-dynamics diagnostic as fig. 6, the final readiness matrix as fig. 20, and the process closure as fig. 21.

Candidate Outcome I

Table 1: Candidate accuracy intervals from `output/data/ml_candidate_intervals.json`; intervals describe the fixed local test split.

Candidate	Status	Correct/test	Accuracy	Wilson 95% interval
baseline	baseline	413/500	82.6%	79.0% to 85.7%
softmax linear	rejected	441/500	88.2%	85.1% to 90.7%
mlp relu 32	rejected	443/500	88.6%	85.5% to 91.1%
tiny patch attention	rejected	152/500	30.4%	26.5% to 34.6%
mlp tanh 64	accepted	447/500	89.4%	86.4% to 91.8%

Candidate Outcome II

Table 2: Candidate rank-stability table from output/data/ml_candidate_rank_stability.json; frequencies are deterministic local bootstrap summaries.

Candidate	Observed rank	Top-rank frequency	Mean rank	Accuracy
softmax linear	3	4.2%	2.554	88.2%
mlp relu 32	2	23.3%	2.130	88.6%
tiny patch attention	4	0.0%	4.000	30.4%
mlp tanh 64	1	72.5%	1.316	89.4%

Run-Derived Figures

Run-Derived Figures I

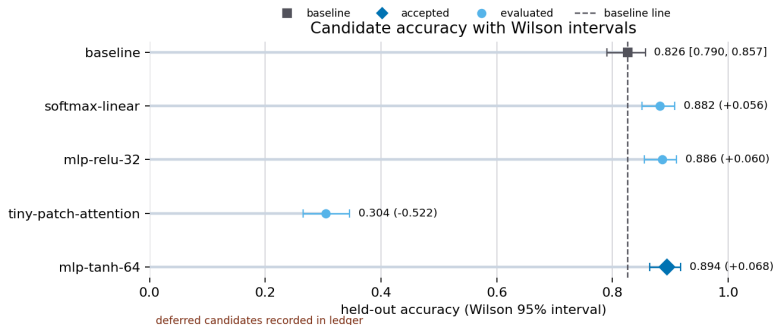


Figure 1: Held-out accuracy with Wilson intervals for the baseline and evaluated candidates from output/data/ml_candidate_intervals.json; accepted candidate exp-mlp-tanh-64 improves accuracy from 82.6% to 89.4% on the fixed subset, with deferred proposals kept in the candidate ledger. Generation method: Lollipop accuracy comparison with Wilson interval error bars and direct labels. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Run-Derived Figures II

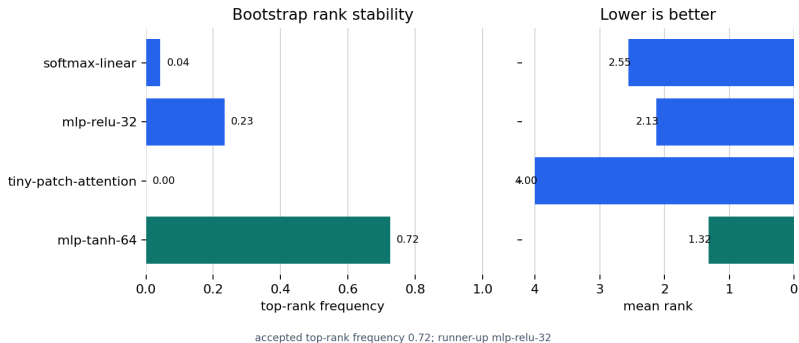
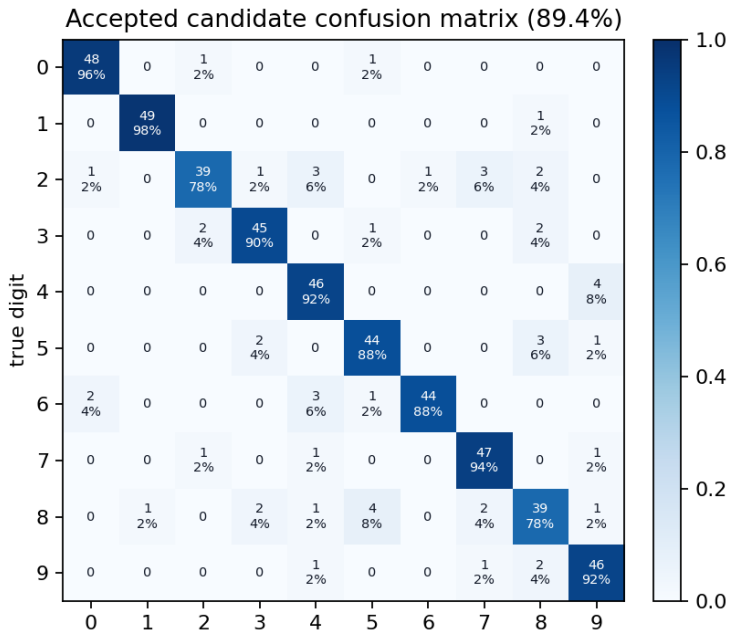


Figure 2: Rank-stability summary for exp-mlp-tanh-64 from output/data/ml_candidate_rank_stability.json; deterministic local bootstrap resampling shows how often each evaluated candidate ranks first under the fixed test split. Generation method: Bootstrap top-rank frequency and mean-rank comparison. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Run-Derived Figures III



Training And Error Diagnostics

Training And Error Diagnostics I

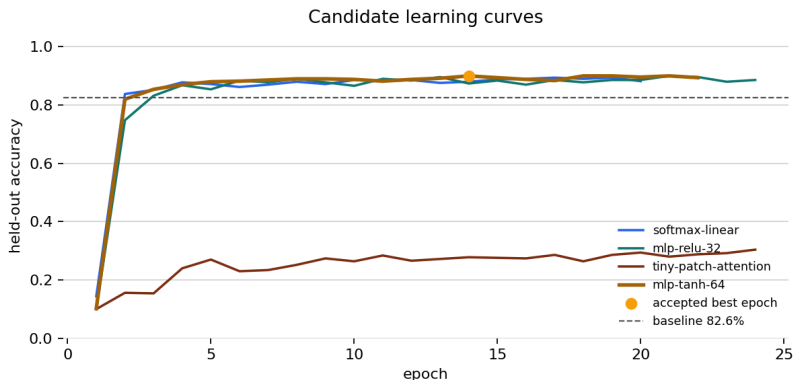


Figure 5: Epoch-level held-out accuracy curves for evaluated candidates from output/data/ml_training_history.csv; the accepted curve is visually emphasized for exp-mlp-tanh-64. Generation method: Epoch-level held-out accuracy lines with accepted best-epoch marker. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics II

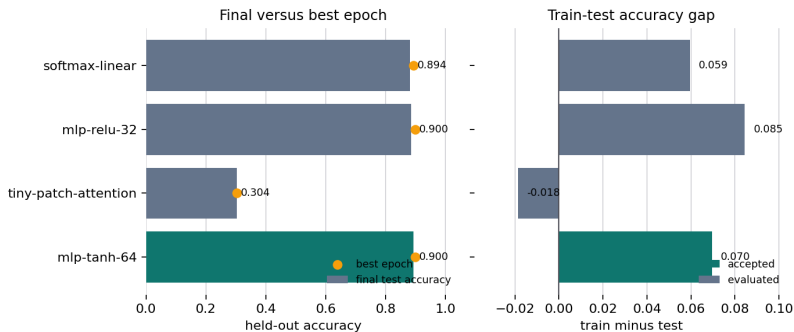


Figure 6: Configured-training dynamics for evaluated candidates from output/data/ml_training_diagnostics.json; exp-mlp-tanh-64 is highlighted while best-epoch markers and train-test gaps remain bounded to the local run. Generation method: Final and best-epoch accuracy bars plus train-test gap bars. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics III

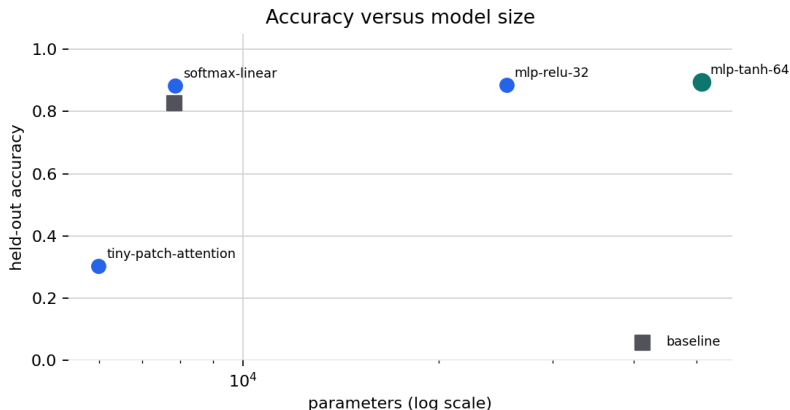


Figure 7: Parameter-count versus held-out accuracy for the baseline and evaluated candidates from `output/data/ml_task_results.json`; the accepted candidate is highlighted without claiming a general scaling law. Generation method: Log-parameter scatter plot against held-out accuracy. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics IV

Accepted candidate error examples

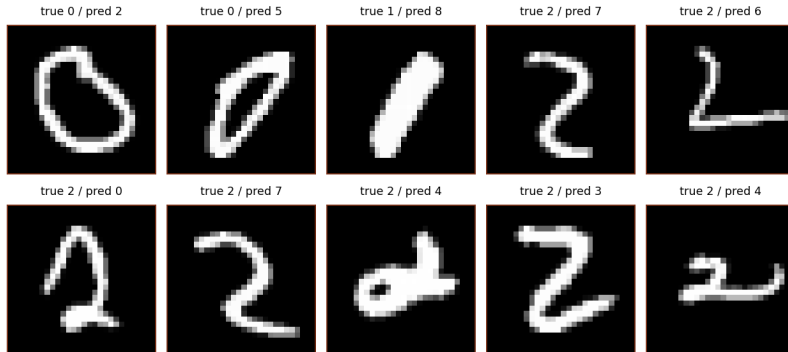


Figure 8: First accepted-candidate error examples for exp-mlp-tanh-64, sourced from output/data/ml_error_examples.json and data/mnist_small.npz; these images support qualitative diagnosis only. Generation method: Deterministic grid of the first accepted-candidate misclassifications. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics V

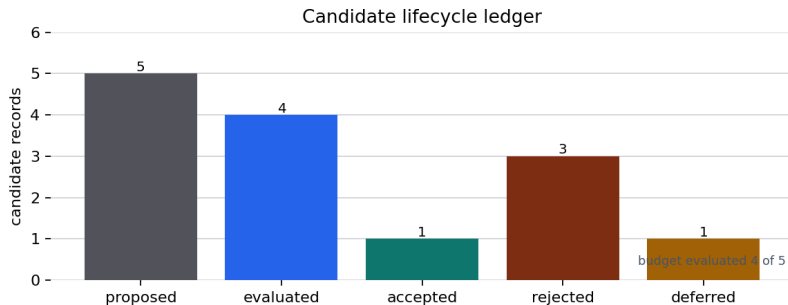


Figure 9: Candidate lifecycle ledger from output/data/ml_candidate_ledger.json: 4 evaluated out of 5 proposed candidates, with deferred proposals kept visible instead of executed automatically. Generation method: Candidate lifecycle status-count bar chart. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics I

The selected candidate's best held-out epoch is 14, its final learning rate is 0.144, its train-loss reduction is 4.161, and its final train-test accuracy gap is 7.0%. These values come from the configured training diagnostics rather than from a manually maintained result summary.

Training And Error Diagnostics I

Table 3: Configured-training diagnostics from `output/data/ml_training_diagnostics.json`.

Candidate	Status	Best epoch	Best test accuracy	Final test accuracy	Train-test gap	Loss reduction	Final learning rate
softmax	rejected	17	89.4%	88.2%	5.9%	2.917	0.273
linear							
mlp relu 32	rejected	21	90.0%	88.6%	8.5%	4.341	0.178
tiny patch	rejected	24	30.4%	30.4%	-1.8%	0.165	0.214
attention							
mlp tanh 64	accepted	14	90.0%	89.4%	7.0%	4.161	0.144

Diagnostic Analysis

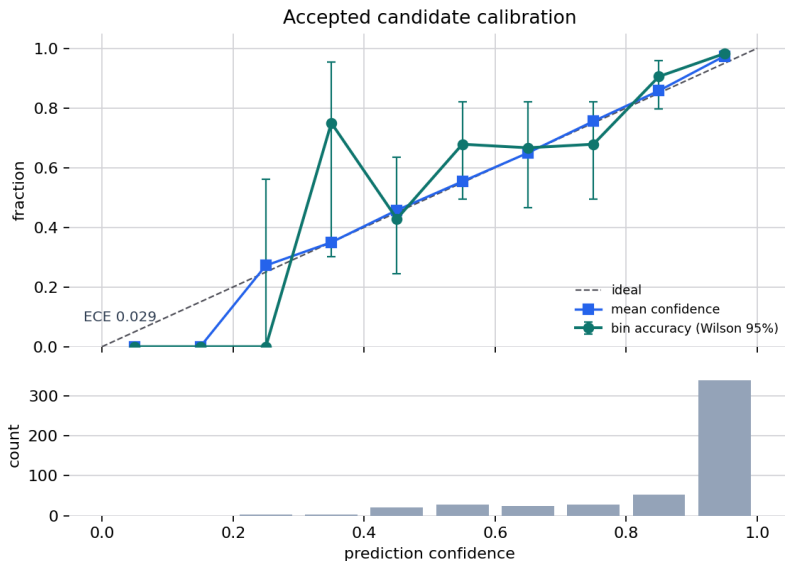
Diagnostic Analysis I

The selected candidate macro F1 is 89.4%, with a held-out accuracy interval of 86.4% to 91.8%. Probability diagnostics report expected calibration error 2.9% and 11 high-confidence error(s). The top non-diagonal confusion pair is 4 $\rightarrow$ 9 (4), and the minimum selected-candidate accuracy across deterministic robustness transforms is 80.0%. These values are descriptive diagnostics for the validated local run, not external benchmark claims.

Classification And Error Structure I

The class-level and error-pattern diagnostics are intentionally visual-first: fig. 10 checks confidence calibration, fig. 11 separates precision, recall, and F1 by digit, and fig. 12 ranks the non-diagonal confusions that most shape the selected-candidate error profile. The accompanying tables preserve the same values for audit and downstream comparison.

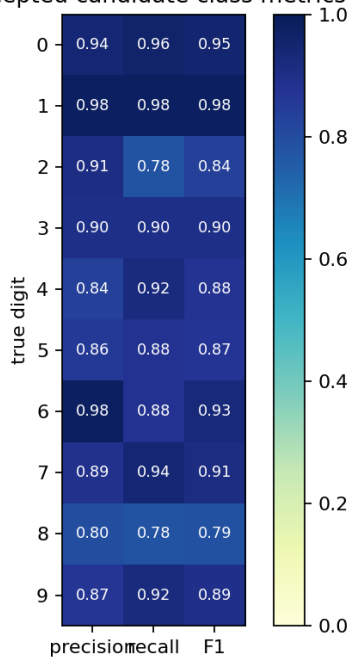
Classification And Error Structure I



Classification And Error Structure II

Figure 10: Reliability curve for exp-mlp-tanh-64 from output/data/ml_calibration_report.json; expected calibration error and bin counts summarize the accepted candidate on the fixed local test split. Generation method: Reliability curve with confidence-bin support histogram. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Accepted candidate class metrics



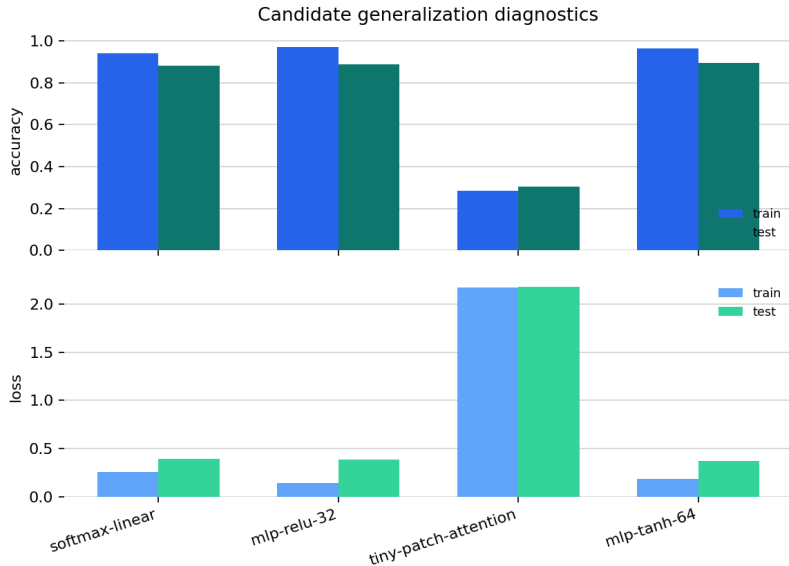
Uncertainty, Generalization, And Perturbation I

Additional uncertainty and matched-comparison diagnostics report bootstrap accuracy interval 86.4% to 92.0%, bootstrap macro-F1 interval 86.5% to 91.9%, paired net accuracy gain 6.8%, exact McNemar p-value 0.000, mean correct-prediction confidence 90.9%, mean error confidence 63.1%, and 22 low-margin selected-candidate prediction(s). These diagnostics are generated from the fixed local test split and are reported as run evidence, not as population-level certification.

Uncertainty, Generalization, And Perturbation I

fig. 13 compares train and test behavior, fig. 14 shows deterministic no-retrain perturbation results, fig. 15 summarizes confidence and margin distributions, fig. 16 shows local resampling intervals, and fig. 17 exposes the matched selected-versus-baseline correctness table used by the paired comparison.

Uncertainty, Generalization, And Perturbation I



Uncertainty, Generalization, And Perturbation II

Figure 13: Train/test accuracy and loss for evaluated candidates from `output/data/ml_classification_diagnostics.json`; the plot exposes local generalization gaps without claiming full-dataset behavior. Generation method: Grouped train/test accuracy and loss bars by evaluated candidate. Registry metadata records the generation method, source artifact, and claim boundary for validation.

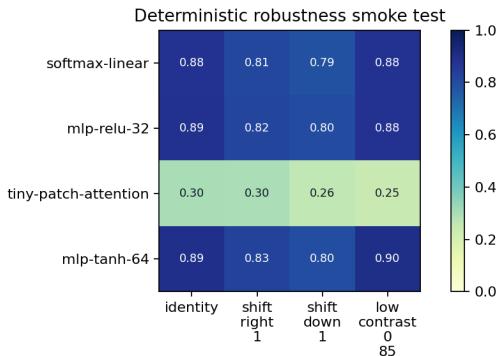
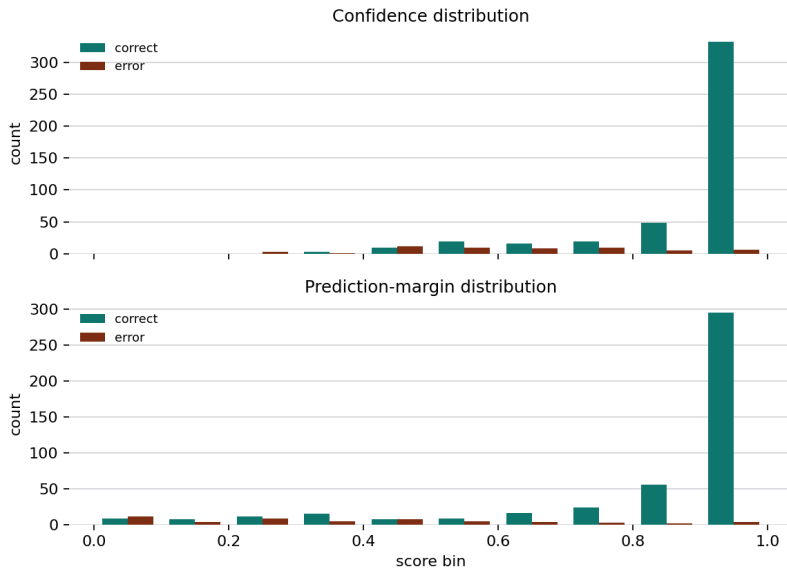


Figure 14: Accuracy for evaluated candidates under identity, one-pixel shifts, and low contrast from `output/data/ml_robustness_report.json`; the deterministic transforms provide a bounded smoke test only. Generation method: Candidate-by-transform accuracy heatmap for deterministic perturbations. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Uncertainty, Generalization, And Perturbation III



Uncertainty, Generalization, And Perturbation IV

Figure 15: Confidence and prediction-margin histograms for exp-mlp-tanh-64 from output/data/ml_probability_diagnostics.json; the figure separates correct and incorrect local test predictions. Generation method: Confidence and margin histograms split by correctness. Registry metadata records the generation method, source artifact, and claim boundary for validation.

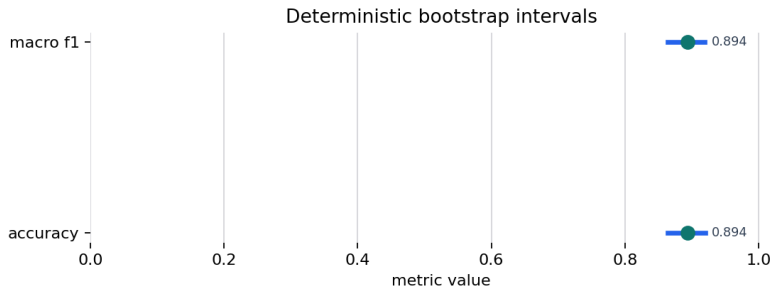
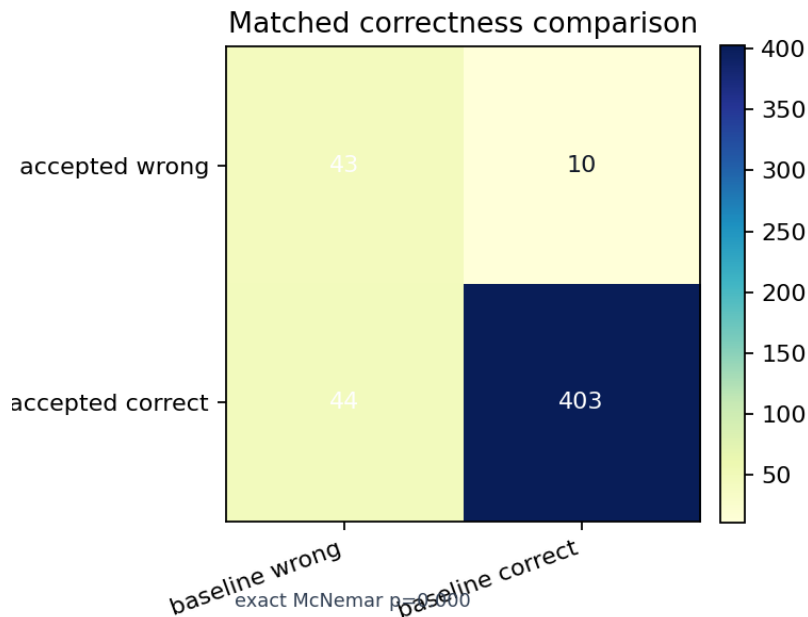


Figure 16: Deterministic percentile-bootstrap intervals for exp-mlp-tanh-64 from output/data/ml_bootstrap_intervals.json; the intervals summarize local sampling variation for accuracy and macro F1. Generation method: Horizontal percentile-bootstrap interval plot. Registry metadata records the generation method, source artifact, and claim boundary for validation.



Probability Quality And Selective Prediction I

Probability-quality diagnostics report Brier score 0.161, negative log likelihood 0.361, top-2 accuracy 95.6%, and Cohen kappa 0.882 for the selected candidate. At the highest configured confidence threshold, retained coverage is 67.8% and selective accuracy is 98.2%.

The selective-prediction view in fig. 18 reports the configured confidence-threshold trade-off: retaining fewer predictions can raise selective accuracy, but the coverage table keeps that trade-off explicit. The candidate probability-quality comparison in fig. 19 keeps accuracy separate from proper-score behavior, so the selected candidate is not treated as automatically best on every diagnostic axis.

Probability Quality And Selective Prediction I

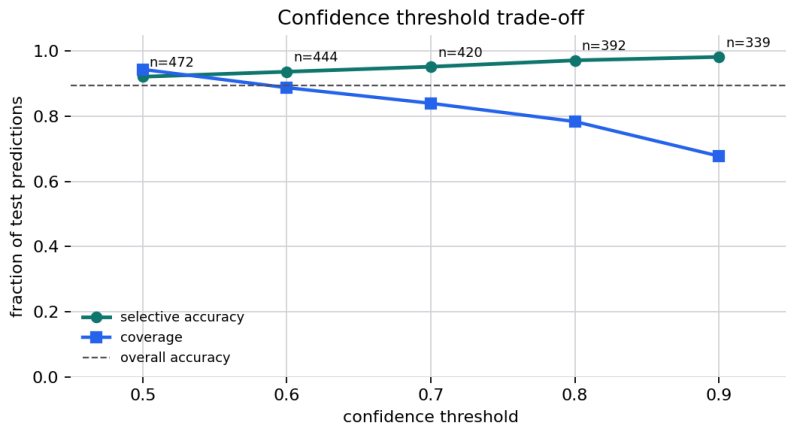


Figure 18: Confidence-threshold trade-off for exp-mlp-tanh-64 from output/data/ml_statistical_summary.json; the plot compares retained coverage, selective accuracy, and the unthresholded accepted-candidate accuracy on the fixed local split. Generation method: Confidence-threshold coverage and selective-accuracy line chart. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Probability Quality And Selective Prediction II

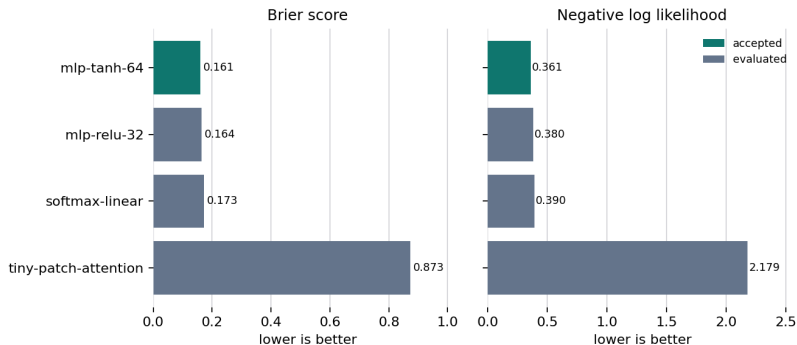


Figure 19: Brier score and negative log likelihood for evaluated candidates from `output/data/ml_statistical_summary.json`; lower values indicate better probability quality within the configured local run, and the accepted candidate is highlighted. Generation method: Brier score and negative-log-likelihood bar comparison. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Probability Quality And Selective Prediction III

Table 12: Accepted-candidate statistical summary from output/data/ml_statistical_summary.json.

Statistic	Value
Accuracy	89.4%
Balanced accuracy	89.4%
Macro F1	89.4%
Top-2 accuracy	95.6%
Cohen kappa	0.882
Brier score	0.161
Negative log likelihood	0.361
Expected calibration error	2.9%

Probability Quality And Selective Prediction IV

Table 13: Selective-accuracy threshold table from output/data/ml_statistical_summary.json.

Confidence threshold	Coverage	Selective accuracy	Retained	Errors
50.0%	94.4%	92.2%	472	37
60.0%	88.8%	93.7%	444	28
70.0%	84.0%	95.2%	420	20
80.0%	78.4%	97.2%	392	11
90.0%	67.8%	98.2%	339	6

Probability Quality And Selective Prediction V

Table 14: Candidate probability-quality table from output/data/ml_statistical_summary.json.

Candidate	Accuracy	Top-2 accuracy	Brier	NLL	Mean confidence
softmax linear	88.2%	95.8%	0.173	0.390	85.1%
mlp relu 32	88.6%	95.6%	0.164	0.380	89.9%
tiny patch	30.4%	46.6%	0.873	2.179	13.1%
attention					
mlp tanh 64	89.4%	95.6%	0.161	0.361	87.9%

Candidate Ledger

Candidate Ledger I

Table 15: Candidate lifecycle ledger from `output/data/ml_candidate_ledger.json`.

Candidate	Model	Status	Test accuracy	Parameters
baseline	nearest-centroid	baseline	82.6%	7840
softmax linear	softmax regression	rejected	88.2%	7850
mlp relu 32	MLP	rejected	88.6%	25450
tiny patch attention	tiny patch-attention	rejected	30.4%	5994
mlp tanh 64	MLP	accepted	89.4%	50890
mlp relu 64 deferred	MLP	deferred	N/A	0

Candidate Ledger I

The candidate-selection audit separates the objective ranking from descriptive diagnostics. It records the configured metric, Wilson interval, probability quality, parameter count, and deterministic tie-break context for each evaluated candidate. The diagnostic-boundary table states what each generated surface supports and what it does not support.

Candidate Ledger I

Table 16: Candidate-selection audit from `output/data/ml_candidate_selection_audit.json`; the objective metric ranks candidates, while probability diagnostics and parameter counts audit the chosen tie-break context.

Rank	Candidate	Status	Accuracy	Wilson 95%	Brier	NLL	Parameters
1	mlp tanh 64	accepted	89.4%	86.4% to 91.8%	0.161	0.361	50890
2	mlp relu 32	rejected	88.6%	85.5% to 91.1%	0.164	0.380	25450
3	softmax linear	rejected	88.2%	85.1% to 90.7%	0.173	0.390	7850
4	tiny patch attention	rejected	30.4%	26.5% to 34.6%	0.873	2.179	5994

Candidate Ledger I

Table 17: Diagnostic claim-boundary table from `output/data/ml_diagnostic_boundary.json`.

Surface	Source	Method	Supports	Does not support
objective selection	task results	rank evaluated candidates by configured held-out metric and deterministic tie...	accepted-candidate selection within this fixed local task	full MNIST state-of-the-art, external benchmark leadership, or universal mode...
descriptive diagnostics	class diagnostics	per-class metrics, calibration, probability quality, and paired comparison	local error analysis and uncertainty description	population-level certification or deployment readiness
robustness smoke test	robustness report	deterministic no-retrain transforms applied to the fixed test split	small perturbation smoke-test evidence	adversarial robustness or distribution-shift robustness
artifact integrity	integrity attestation	local SHA-256 checks over declared inputs and generated artifacts	local artifact integrity evidence for the run	external signing, production SLSA compliance, or runtime intrusion detection
review governance	review decisions	deferred generated review decisions with human review required	readiness for human review	machine self-approval or publication acceptance

Readiness And Review Artifacts

Readiness And Review Artifacts I

The broader AutoResearch run writes the reproducibility, benchmark, review, and manuscript-hydration surfaces summarized below. The schema manifest records 31 schema-versioned governance payload(s), and the local research-object manifest records 84 observed artifact record(s) with checksums and approval state false. The phase ledger records 3 settlement pass(es), while the figure-quality report covers 25 registered figure(s) with validity false.

Readiness And Review Artifacts I

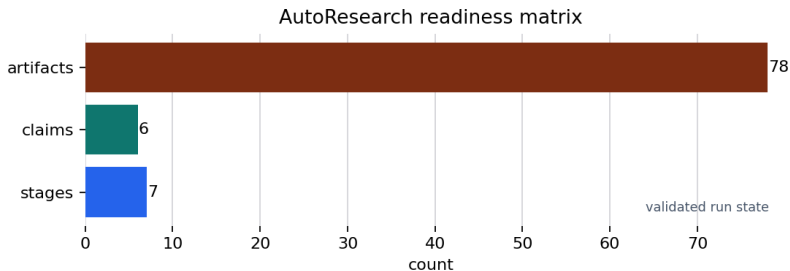


Figure 20: Validated AutoResearch run with 7 stages, 6 supported claims, and 78 required artifacts from output/data/autoresearch_loop.json; the count summarizes readiness artifacts, not human approval. Generation method: Horizontal count summary from final loop metrics. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Readiness And Review Artifacts II

File-backed AutoResearch closure



Figure 21: File-backed AutoResearch closure from program through review, with 6 supported claims and readiness passed; review remains a deferred human decision and the provenance path remains inspectable. Generation method: File-backed process-flow diagram from final loop state. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Table 18: Generated artifact manifest from output/reports/artifact_manifest.json.

Artifact	Role	Bytes
output/data/autoresearch_claims.json	Loop artifact	1766
output/data/autoresearch_evidence_registry	Evidence registry	4436
output/data/autoresearch_integrity_artifacts_registry	Integrity evidence	22755
output/data/autoresearch_inventory_registry	Inventory evidence	20197
output/data/autoresearch_loop.json	Loop artifact	16085
output/data/autoresearch_phase_ledger	Loop candidate ledger	3779
output/data/autoresearch_plan.json	Loop artifact	19818
output/data/autoresearch_review_packet	Review packet	16204
output/data/autoresearch_schema_manifest	Manifest artifact	7226
output/data/autoresearch_security_proof	Security evidence	1537
output/data/autoresearch_stage_manifest	Loop artifact	749
output/data/autoresearch_supply_chain_inventory	Inventory evidence	21453
output/data/autoresearch_threat_model	Security evidence	6370
output/data/benchmark_boundary.json	Benchmark grading	2363
output/data/benchmark_scores.json	Benchmark grading	621

Readiness And Review Artifacts III

output/data/figure_quality_report.json	Loop artifact	16041
output/data/figure_style.json	Loop artifact	1117
output/data/idea_ledger.json	Run or candidate ledger	5233
output/data/manuscript_figure_blocks.json	Manuscript hydration	13062
output/data/manuscript_variable_provider.json	Manuscript hydration	30891
output/data/manuscript_variables.json	Manuscript hydration	57706
output/data/ml_bootstrap_intervals.json	Loop artifact	615
output/data/ml_calibration_bin_intervals.json	Loop artifact	2879
output/data/ml_calibration_report.json	Loop artifact	2201
output/data/ml_candidate_intervals.json	Loop artifact	1577
output/data/ml_candidate_ledger.json	Run or candidate ledger	570872
output/data/ml_candidate_rank_stability.json	Loop artifact	3135
output/data/ml_candidate_selection_history.json	Loop artifact	2145
output/data/ml_class_balance.json	Loop artifact	2393
output/data/ml_classification_diagnostic.json	Loop artifact	4246
output/data/ml_confusion_matrix.csv	Loop artifact	271
output/data/ml_diagnostic_boundary.json	Loop artifact	2064
output/data/ml_error_examples.json	Loop artifact	989
output/data/ml_paired_comparison.json	Loop artifact	470
output/data/ml_prediction_records.json	Loop artifact	989913
output/data/ml_probability_diagnostic.json	Loop artifact	3045
output/data/ml_robustness_report.json	Loop artifact	3758
output/data/ml_statistical_summary.json	Loop artifact	3032
output/data/ml_task_results.json	Loop artifact	703566
output/data/ml_training_diagnostics.json	Loop artifact	2968
output/data/ml_training_history.csv	Loop artifact	6775
output/data/mnist_task_config.json	Loop artifact	3926
output/data/publication_ledger.json	Run or candidate ledger	1005
output/data/research_object_manifest.json	Loop artifact	20789
output/data/research_program.json	Loop artifact	965
output/data/review_decisions.json	Review packet	669
output/data/run_ledger.json	Run or candidate ledger	328
output/data/transmission_manifest.json	Loop artifact	503
output/figures/autoresearch_candidate_generated_figure.png	Generated figure	28056
output/figures/autoresearch_closure_generated_figure.png	Generated figure	40873
output/figures/autoresearch_integrity_generated_figure.png	Generated figure	48228
output/figures/autoresearch_security_generated_figure.png	Generated figure	86494

Readiness And Review Artifacts IV

output/figures/autoresearch_stage_machine.png	Generated figure	23486
output/figures/figure_registry.json	Loop artifact	30674
output/figures/ml_bootstrap_intervals.png	Generated figure	21830
output/figures/ml_calibration_reliability.png	Generated figure	73855
output/figures/ml_candidate_rank_statistics.png	Generated figure	43670
output/figures/ml_candidate_scores.png	Generated figure	59947
output/figures/ml_classification_metrics_heatmap.png	Generated figure	55133
output/figures/ml_complexity_accuracy.png	Generated figure	35016
output/figures/ml_confusion_matrix.png	Generated figure	65918
output/figures/ml_confusion_pairs.png	Generated figure	33982
output/figures/ml_generalization_gap.png	Generated figure	48554
output/figures/ml_learning_curves.png	Generated figure	59360
output/figures/ml_paired_correctness.png	Generated figure	43306
output/figures/ml_per_class_accuracy.png	Generated figure	34938
output/figures/ml_probability_margin_distribution.png	Generated figure	41747
output/figures/ml_probability_quality.png	Generated figure	38072
output/figures/ml_robustness_matrix.png	Generated figure	51292
output/figures/ml_selective_accuracy.png	Generated figure	48332
output/figures/ml_training_dynamics.png	Generated figure	50908
output/figures/mnist_class_balance.png	Generated figure	27006
output/figures/mnist_error_examples.png	Generated figure	28446
output/figures/mnist_subset_contact_sheet.png	Generated figure	23682
output/figures/transmission_integrity_comparison.png	Generated figure	18338
output/figures/transmission_pairing.png	Generated figure	11314
output/manuscript/00_abstract.md	Manuscript hydration	1451
output/manuscript/01_introduction.md	Manuscript hydration	14774
output/manuscript/02_methodology.md	Manuscript hydration	21101
output/manuscript/03_results.md	Manuscript hydration	42052
output/manuscript/04_conclusion.md	Manuscript hydration	2642
output/manuscript/99_references.md	Manuscript hydration	58
output/manuscript/config.yaml	Manuscript hydration	3233
output/manuscript/preamble.md	Manuscript hydration	137
output/manuscript/references.bib	Manuscript hydration	23228
output/pdf/template_autoresearch_project_article.pdf	Loop artifact	1463957
output/release/sw_h_repo_url.txt	Loop artifact	59
output/release/upload_receipts.json	Loop artifact	2004
output/reports/autoresearch_evidence_evidence_registry	Evidence registry	1136

Readiness And Review Artifacts V

output/reports/autoresearch_loop.json	Loop artifact	16085
output/reports/autoresearch_loop.md	Loop artifact	1982
output/reports/autoresearch_review_packet.md	Review artifact	912
output/reports/autoresearch_security_review_packet.md	Review artifact	1104
output/reports/autoresearch_summary_loop	Loop artifact	255
output/reports/benchmark_readiness_benchmark.json	Benchmark grading	778
output/reports/manuscript_composition_manuscript	Manuscript hydration	2149
output/reports/ml_benchmark_score.json	Benchmark grading	382
output/reports/ml_experiment_report_loop	Loop artifact	1687
output/reports/test_results.json	Loop artifact	13815
output/reports/test_results.md	Loop artifact	292
output/slides/00_abstract_slides.pdf	Loop artifact	19958
output/slides/01_introduction_slides.pdf	Loop artifact	64117
output/slides/02_methodology_slides.pdf	Loop artifact	241270
output/slides/03_results_slides.pdf	Loop artifact	1140252
output/slides/04_conclusion_slides.pdf	Loop artifact	21552
output/slides/99_references_slides.pdf	Loop artifact	11244
output/web/_combined_manuscript.manuscript	Manuscript hydration	83966
output/web/favicon.ico	Loop artifact	121
output/web/index.html	Loop artifact	164706
output/web/manuscript__00_abstract.manuscript	Manuscript hydration	10237
output/web/manuscript__01_introduction.manuscript	Manuscript hydration	23638
output/web/manuscript__02_methodology.manuscript	Manuscript hydration	37355
output/web/manuscript__03_results.manuscript	Manuscript hydration	78026
output/web/manuscript__04_conclusion.manuscript	Manuscript hydration	11184
output/web/manuscript__99_references.manuscript	Manuscript hydration	8552

Readiness And Review Artifacts VI

Table 19: Review-gate decisions from `output/data/review_decisions.json`.

Gate	Required	Decision	Rationale
proposal_review	True	deferred	Decision is read from <code>human_review.yaml</code> when present; generated readiness is not approval.
evidence_review	True	deferred	Decision is read from <code>human_review.yaml</code> when present; generated readiness is not approval.

Readiness And Review Artifacts VII

Table 20: Benchmark grading table from output/data/benchmark_scores.json.

Benchmark task	Status	Score	Grading output
readiness-smoke	graded	1	output/reports/benchmark_readiness_
ml-loop-score	graded	1	output/reports/ml_benchmark_score.j

Table 21: Deterministic phase ledger from output/data/autoresearch_phase_ledger.json; settlement order is not an autonomy claim.

Phase	Order	Group	Observed artifacts	Description
intrinsic readiness	1	readiness	3	validate configured project-intrinsic contracts
core artifacts	2	loop	0	write plan, stage matrix, and provisional loop outputs
evidence registry	3	evidence	2	write local evidence registry
ml task	4	ml	54	run fixed-seed bounded candidate evaluation
method contract	5	governance	3	write program, idea, run, review, and benchmark ledgers
provisional payloads	6	settlement	12	refresh loop payloads before extrinsic validation
security artifacts	7	security	10	write local security and integrity evidence
final visuals	8	figures	54	write final registry-backed figures
manuscript hydration	9	manuscript	9	write variables, provenance, and figure blocks
readiness manifest	10	settlement	12	refresh checksum manifest before extrinsic validation
schema manifest	11	schema	1	write generated JSON schema-version manifest
research object manifest	12	packaging	1	write local research-object manifest

Readiness And Review Artifacts IX

extrinsic readiness	13	readiness	3	validate generated artifacts and extrinsic contracts
final schema manifest	14	schema	1	refresh schema manifest after final payload updates
final research object manifest	15	packaging	1	refresh local research-object manifest
artifact manifest	16	settlement	12	write final artifact checksum manifest

Readiness And Review Artifacts X

Table 22: Figure-quality checks from output/data/figure_quality_report.json; 25 registered figure(s) were checked.

Figure	Pixels	Variance	Source exists	Nonblank
fig:autoresearch_candidate_cycle	1184x480	0.082	True	True
fig:autoresearch_closure	666x448	0.015	True	True
fig:autoresearch_integrity	440x736	0.040	True	True
fig:autoresearch_security_matrix	472x736	0.014	True	True
fig:autoresearch_stage_1	1120x416	0.079	True	True
fig:ml_bootstrap_inter	1152x448	0.009	True	True
fig:ml_calibration_reliability	1152x832	0.015	True	True
fig:ml_candidate_rank	1408x560	0.053	True	True
fig:ml_candidate_score	1376x688	0.013	True	True
fig:ml_classification_metrics	928x832	0.092	True	True
fig:ml_complexity_accuracy	1120x608	0.010	True	True
fig:ml_confusion_matrix	896x768	0.045	True	True
fig:ml_confusion_pairs	1152x576	0.129	True	True
fig:ml_generalization_gap	1184x864	0.059	True	True
fig:ml_learning_curves	1216x608	0.018	True	True
fig:ml_paired_correctness	1152x672	0.075	True	True
fig:ml_per_class_accuracy	1152x512	0.098	True	True
fig:ml_probability_margin_distribution	1184x864	0.024	True	True
fig:ml_probability_qualities	1134x608	0.046	True	True
fig:ml_robustness_matrix	1280x608	0.073	True	True
fig:ml_selective_accuracy	1088x608	0.016	True	True
fig:ml_training_dynamics	1408x608	0.072	True	True
fig:mnist_class_balanced	216x544	0.071	True	True
fig:mnist_error_examples	1280x736	0.234	True	True
fig:mnist_subset_contain	1216x544	0.230	True	True

Security Readiness And Integrity Evidence

Security Readiness And Integrity Evidence I

The local security profile reports attestation status passed after checking 80 file record(s), with 0 missing record(s) and 0 checksum mismatch(es). The inventory contains 14 input record(s) and 69 generated-artifact record(s). The integrity-chain figure is fig. 22. These values support local artifact-integrity claims only; they do not claim external signing, production SLSA compliance, or runtime security monitoring.

Security Readiness And Integrity Evidence I

Local artifact integrity chain

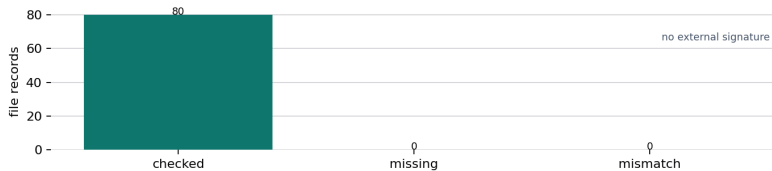


Figure 22: Local integrity chain from output/data/autoresearch_integrity_attestation.json; checksums summarize the observed run artifacts and remain unsigned local evidence. Generation method: Local checksum attestation chain with checked, missing, and mismatch counts. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Table 23: Integrity-attestation summary from output/data/autoresearch_integrity_attestation.json.

Integrity field	Value
status	passed
algorithm	sha256
checked files	80
missing files	0
checksum mismatches	0
external signature	False

Manuscript Hydration Provenance

The final run supports 6 manuscript-facing claim(s) and checks 78 required artifact(s). The rendered manuscript uses injected variables from generated data payloads, so the abstract and results track the latest analysis run rather than hard-coded counts. The final readiness status is passed; generated review decisions are recorded as deferred for human review rather than as self-approval.

Manuscript Hydration Provenance I

Table 24: Variable provenance summary from output/data/manuscript_variable_provenance.json.

Source artifact	Injected variables or fragments
output/data/autoresearch_integrity_attestation.json	5
output/data/autoresearch_loop.json	57
output/data/autoresearch_phase_ledger.json	2
output/data/autoresearch_schema_manifest.json	1
output/data/autoresearch_security_profile.json	6
output/data/autoresearch_supply_chain_inventory.json	3
output/data/autoresearch_threat_model.json	4
output/data/benchmark_scores.json	1
output/data/figure_quality_report.json	3
output/data/manuscript_variable_provenance.json	1
output/data/ml_bootstrap_intervals.json	3
output/data/ml_calibration_bin_intervals.json	1
output/data/ml_calibration_report.json	3
output/data/ml_candidate_intervals.json	1
output/data/ml_candidate_ledger.json	1
output/data/ml_candidate_rank_stability.json	3
output/data/ml_candidate_selection_audit.json	1
output/data/ml_class_balance.json	3
output/data/ml_classification_diagnostics.json	5
output/data/ml_diagnostic_boundary.json	1
output/data/ml_paired_comparison.json	3
output/data/ml_probability_diagnostics.json	4
output/data/ml_robustness_report.json	2
output/data/ml_statistical_summary.json	9
output/data/ml_task_results.json	39
output/data/ml_training_diagnostics.json	7
output/data/research_object_manifest.json	2
output/data/review_decisions.json	1
output/figures/figure_registry.json	51
output/reports/artifact_manifest.json	1